



PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Blood Bank Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Team 9	04/09/2026	-	1.0
Team 9	02/10/2026	Non-Functional Requirements, Appendix Update	1.0
Team 9	04/10/2026	Updated functional, security, and traceability requirements for consistency with STP and SAD.	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to define the functional and non-functional requirements of the Blood Bank Management System. The system is intended to support donor management, donation recording, blood inventory tracking, blood availability search, blood request processing, blood issuance, and reporting.

1.2 Intended Audience

This document is intended for the project team, developers, testers, faculty evaluators, and other users involved in the development and review of the Blood Bank Management System. It provides a common understanding of the system requirements and serves as a reference during the design, implementation, and testing stages of the project.

1.3 Product Scope

The Blood Bank Management System is a web-based application developed to manage blood donation and blood availability within a blood bank. The system will maintain donor details and donation history, record blood donations, manage blood stock, monitor expired or low-stock blood units, and allow users to search for required blood groups. It will also support blood requests, approval and fulfilment of requests, and generation of basic reports related to donors, donations, inventory, and issued blood.

The main objective of the system is to maintain blood-bank information in an organized manner and make it easier for authorized users to track donations, available stock, and blood requests.

1.4 References

- [1] IEEE Std 29148-2018, Systems and Software Engineering - Requirements Engineering.*
- [2] Blood Bank Management System -Project Requirements and Scope, Version 1.0, 2026.*
- [3] OWASP, Application Security Verification Standard.*
- [4] Django Software Foundation, Django Documentation.*

2. Overall Description

2.1 Product Perspective

The Blood Bank Management System is a web-based, stand-alone application that supports the day-to-day operations of a blood bank. It combines donor records, donation details, blood inventory, blood requests, blood issuance, and reporting into one system.

The idea is to eliminate the need for separate sheets or manual records and have all the related information in one place. Blood bank staff can record blood donations and manage stock. Authorized users can check blood availability and process blood requests.

The system is designed to work independently and does not require incorporation into a pre-existing hospital management system. Its modularity should allow additional services or integrations in the future

2.2 Product Functions

The Blood Bank Management System provides the following major functions:

- *Authenticate authorized users and control access according to assigned roles.*
- *Allow administrators to create authorized user accounts and manage user roles.*
- *Register donors and maintain donor information and donation history.*
- *Record blood donations, including donation date, blood group, quantity, and status.*
- *Maintain available blood stock according to blood group.*
- *Monitor expired and near-expiry blood units and identify low-stock blood groups.*
- *Search for the availability of a required blood group.*
- *Create blood requests and track their status.*

- Approve or reject blood requests based on availability and authorization.
- Issue blood against approved requests and update the inventory accordingly.
- Maintain a history of donations, requests, and issued blood.
- Generate basic reports related to donors, donations, inventory, blood requests, and blood issuance.

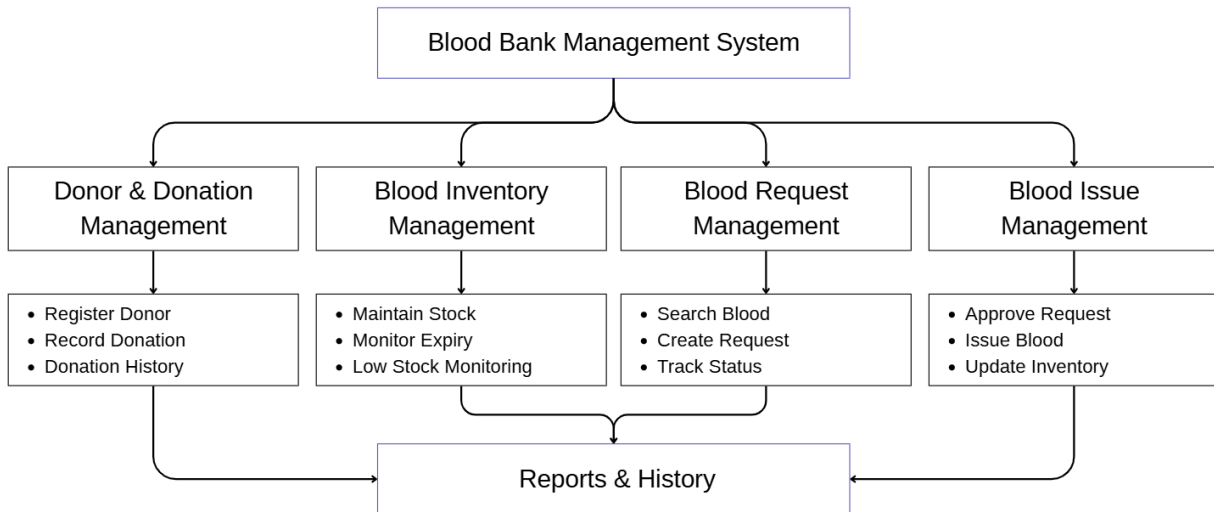


Figure 1: High-Level Functional View of the Blood Bank Management System

2.3 User Classes and Characteristics

Blood Bank Staff

Blood bank staff are the primary operational users of the system. They register donors, record blood donations, maintain blood inventory, monitor expiry and low-stock information, process blood requests, and record blood issuance. These users are expected to have basic computer knowledge and familiarity with blood bank procedures.

Hospital Organization Staff

Authorized staff from hospitals or other requesting organizations use the system to search for the availability of required blood groups, submit blood requests, and track the status of those requests. Their access is limited to blood search and request-related functions.

Administrator

The administrator manages authorized users and oversees the records maintained in the system. The administrator can review donor, donation, inventory, request, and blood-issue information and access system reports. Administrative functions are restricted to users with the required privileges.

2.4 Operating Environment

The Blood Bank Management System will operate as a web-based application accessible through commonly used web browsers such as Google Chrome, Microsoft Edge, and Firefox. The application will be hosted on a server running the backend application and database. Users will access the system from desktop or laptop computers connected to the network or internet.

The proposed implementation will use Python with Django for the application backend and a relational database for storing donor, donation, inventory, request, and blood-issue records. The system will be

accessible through modern web browsers on Windows, macOS, and Linux-based systems. The server-side application may be developed and deployed on a supported Windows or Linux environment.

2.5 Design and Implementation Constraints

The system will be developed as a web-based application using Python and Django. A relational database will be used to store donor, donation, inventory, blood request, and blood issue records.

The application will follow a client-server architecture and will be accessed through standard web browsers. Communication between the browser and server will use HTTP/HTTPS.

Access to system functions will be controlled according to user roles. Functions such as inventory modification, request approval, and blood issuance will be restricted to authorized users.

The first version of the system will operate independently and will not require integration with external hospital systems, live location services, or third-party medical databases.

The implementation should follow consistent coding and design practices so that the system can be maintained and extended in later versions.

2.6 Assumptions and Dependencies

Assumptions

The system assumes that donor, donation, inventory, and blood-request information entered by authorized users is accurate and complete.

Blood bank staff are assumed to verify donor details and the status of donated blood before it is added to the available inventory.

Users are assumed to have access to a supported web browser and a stable network connection while using the application.

The system assumes that medical screening, blood testing, and other physical procedures required before accepting donated blood are performed outside the software system.

Dependencies

The application depends on the availability of the web server and database server for normal operation.

The system depends on Python, the Django framework, and the selected relational database management system for its implementation and operation.

The client application depends on a modern web browser capable of supporting standard web technologies.

Version 1.0 does not depend on external hospital management systems, live location services, or third-party blood bank databases.

3. External Interface Requirements

3.1 User Interfaces

The Blood Bank Management System will provide a browser-based graphical user interface. The interface will be designed to provide clear and consistent access to system functions based on the role of the logged-in user.

Blood bank staff will be able to access donor registration, donation recording, inventory management, blood request processing, and blood issuance functions. Hospital or requesting organization users will be able to search blood availability, submit blood requests, and track request status. Administrators will have access to user management, reports, and other administrative functions.

The major user interfaces will include:

- *Login screen*
- *Donor registration and donor details screen*
- *Donation recording screen*
- *Blood inventory screen*

- *Blood availability search screen*
- *Blood request and request-status screen*
- *Request approval and blood issue screen*
- *Reports and history screen*
- *Administrator user-management screen*

Input forms will clearly identify required fields and display suitable validation messages for missing or invalid data. Common actions such as Save, Update, Search, Approve, Reject, Cancel, and Logout will use consistent labels throughout the application.

3.2 Software Interfaces

The Blood Bank Management System will be developed using Python with the Django web framework. A relational database management system will be used to store information related to donors, donations, blood inventory, blood requests, blood issuance, users, and reports.

The application will communicate with the database to store, retrieve, and update system records. Data exchanged between the application and database will include donor details, donation records, inventory information, blood request details, issue records, and user information.

Users will access the system through a standard web browser. The browser will send user input and requests to the application server, and the server will return the required pages, search results, validation messages, and status information.

3.3 Communications Interfaces

The Blood Bank Management System will be accessed through a web browser. Communication between the browser and the application server will take place using HTTP/HTTPS.

User input submitted through web forms will be sent to the server for processing, and the server will return the required pages, search results, validation messages, and status information.

HTTPS will be used during deployment to protect data transmitted between the client and server. The current version of the system does not require e-mail, SMS, FTP, or other external communication services.

3.4 Hardware Interfaces

The Blood Bank Management System requires no specialized hardware interfaces. Users will access the system through a desktop or laptop computer with a supported web browser and network connection.

4. Analysis Models

4.1 Use Case Diagram

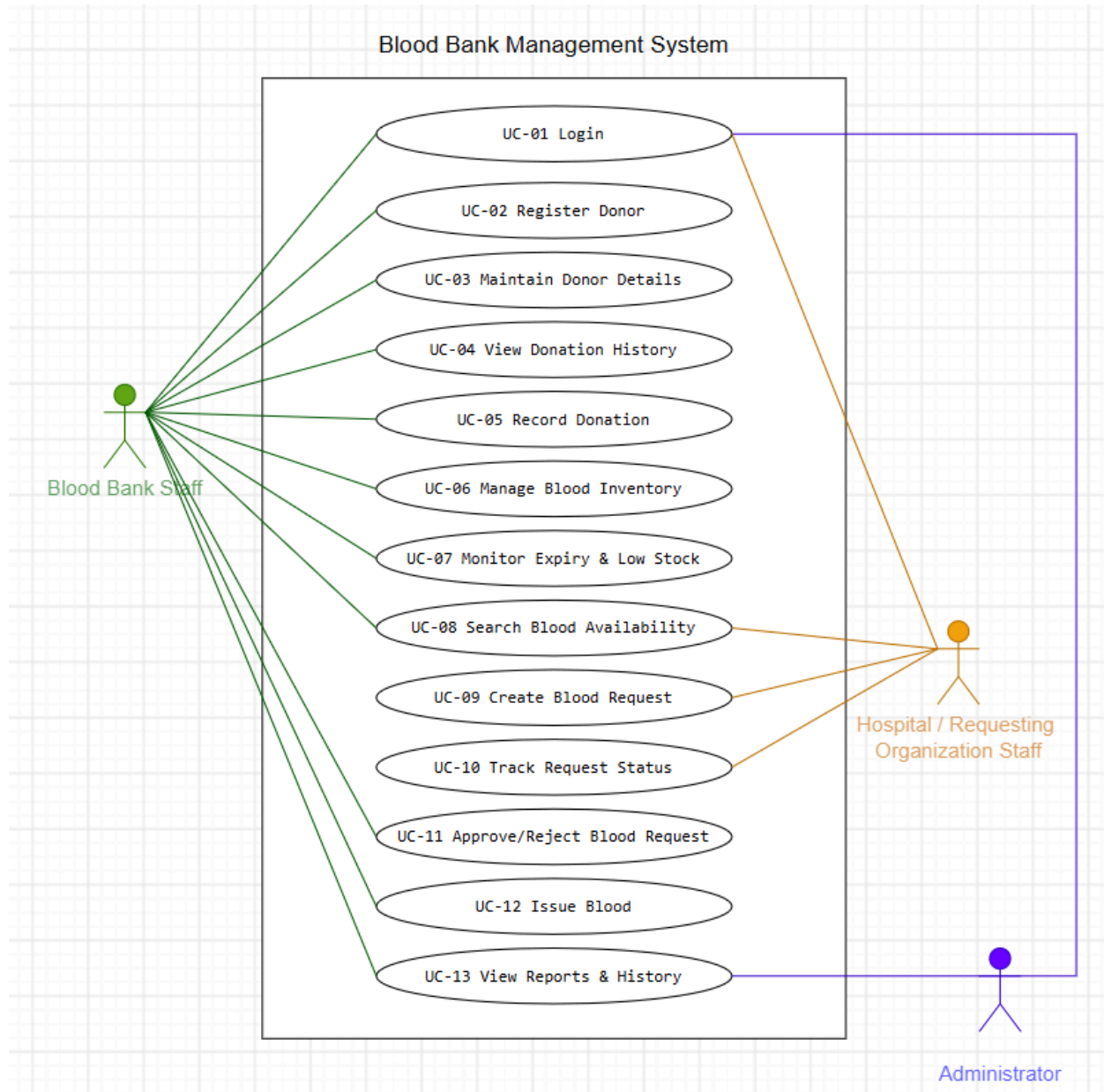


Figure 2: Use Case Diagram of the Blood Bank Management System

5. System Features

5.1 Donor Registration and Management

5.1.1 Description and Priority

This feature allows authorized blood bank staff to register new donors and maintain donor information. It also provides access to a donor's previous donation records when required.

Priority: High

5.1.2 Stimulus/Response Sequences

1. Blood bank staff selects the option to register a donor.
2. The system displays the donor registration form.
3. The staff member enters the required donor information.
4. The system validates the entered information.
5. If the information is valid, the system stores the donor record and confirms successful registration.
6. If required information is missing or invalid, the system displays an appropriate validation message.
7. Staff can later search for an existing donor and update the donor's details.

5.1.3 Functional Requirements

FR-1.1: The system shall allow authorized blood bank staff to register a new donor.

FR-1.2: The system shall require all mandatory donor details before completing registration.

FR-1.3: The system shall store the donor's blood group along with the donor record.

FR-1.4: The system shall allow authorized staff to search for an existing donor.

FR-1.5: The system shall allow authorized staff to update donor information.

FR-1.6: The system shall display the donor's recorded donation history.

FR-1.7: The system shall display a validation message when required donor information is missing or invalid

5.2 Donation Recording

5.2.1 Description and Priority

This feature allows authorized blood bank staff to record donations made by registered donors. The donation record includes relevant information such as the donor, blood group, donation date, quantity, and status.

Priority: High

5.2.2 Stimulus/Response Sequences

1. Staff searches for and selects a registered donor.
2. Staff selects the option to record a donation.
3. The system displays the donation form.
4. Staff enters the donation details.
5. The system validates the entered information.
6. The system stores the donation record.
7. An accepted donation is made available for inventory management.
8. The system confirms that the donation has been recorded.

5.2.3 Functional Requirements

FR-2.1: The system shall allow authorized staff to record a blood donation for a registered donor.

FR-2.2: The system shall record the date of donation.

FR-2.3: The system shall record the blood group associated with the donation.

FR-2.4: The system shall record the quantity or number of blood units donated.

FR-2.5: The system shall maintain the status of each donation as Pending, Accepted, or Rejected.

FR-2.6: The system shall add the donation to the donor's donation history.

FR-2.7: The system shall reject the donation record if required information is missing or invalid.

5.3 Blood Inventory Management

5.3.1 Description and Priority

This feature maintains the available blood stock according to blood group and blood units. Inventory information is updated when accepted blood is added or when blood is issued.

Priority: High

5.3.2 Stimulus/Response Sequences

1. Authorized staff open the blood inventory.
2. The system displays the currently available blood stock.
3. Staff may view stock according to blood group.
4. Accepted blood units are added to the inventory.
5. Issued blood units are deducted from the inventory.
6. The system displays the updated stock.

5.3.3 Functional Requirements

FR-3.1: The system shall maintain the available quantity of blood for each blood group.

FR-3.2: The system shall add accepted blood units to the corresponding blood-group inventory.

FR-3.3: The system shall reduce the available inventory when blood is issued.

FR-3.4: The system shall allow authorized staff to view current inventory information.

FR-3.5: The system shall prevent the available inventory quantity from becoming negative.

5.4 Expiry and Stock Monitoring

5.4.1 Description and Priority

This feature helps blood bank staff monitor blood stock by identifying expired or near-expiry blood units and blood groups with low available stock.

Priority: High

5.4.2 Stimulus/Response Sequences

1. Staff opens the inventory monitoring section.
2. The system checks the stored blood-unit information.
3. The system identifies units that have expired or are approaching expiry.
4. The system identifies blood groups with low available stock.
5. Staff can view the resulting information.

5.4.3 Functional Requirements

FR-4.1: The system shall maintain expiry information for all stored blood units.

FR-4.2: The system shall identify blood units that have expired.

FR-4.3: The system shall identify blood units as near-expiry when they are due to expire within 7 days.

FR-4.4: The system shall identify a blood group as low stock when its available quantity falls below 5 units.

FR-4.5: Expired blood units shall not be considered available for issue.

5.5 Blood Availability Search

5.5.1 Description and Priority

This feature allows authorized users to search the current inventory for a required blood group and view its availability.

Priority: High

5.5.2 Stimulus/Response Sequences

1. *The user opens the blood availability search.*
2. *The user selects or enters the required blood group.*
3. *The system searches the current inventory.*
4. *The system displays the availability and available quantity.*
5. *If no suitable blood is available, the system informs the user.*

5.5.3 Functional Requirements

FR-5.1: *The system shall allow authorized users to search for blood by blood group.*

FR-5.2: *The system shall display the current available quantity for the selected blood group.*

FR-5.3: *The system shall exclude expired blood units from availability results.*

FR-5.4: *The system shall indicate when the requested blood group is unavailable.*

5.6 Blood Request Management

5.6.1 Description and Priority

This feature allows an authorized hospital or requesting organization to submit a request for blood and track its status.

Priority: High

5.6.2 Stimulus/Response Sequences

1. *An authorized requesting user opens the blood request form.*
2. *The user enters the required blood group, quantity, and request details.*
3. *The system validates the request.*
4. *The system creates the request with a pending status.*
5. *Authorized blood bank staff reviews the request.*
6. *The request status is updated as it is processed.*
7. *The requesting user can view the current request status.*

5.6.3 Functional Requirements

FR-6.1: *The system shall allow an authorized requesting organization to submit a blood request.*

FR-6.2: *Each request shall contain the required blood group and quantity.*

FR-6.3: *The system shall assign a status to each blood request.*

FR-6.4: *The system shall initially record a valid new request as pending.*

FR-6.5: *The system shall allow authorized blood bank staff to review submitted requests.*

FR-6.6: *The system shall allow the requesting user to view the current status of a submitted request.*

FR-6.7: *The system shall reject submission when mandatory request information is missing.*

5.7 Blood Issue and Fulfilment

5.7.1 Description and Priority

This feature allows authorized blood bank staff to approve or reject blood requests and issue available blood against an approved request.

Priority: High

5.7.2 Stimulus/Response Sequences

1. *Staff selects a pending blood request.*
2. *The system displays the request details and relevant blood availability.*
3. *Staff approves or rejects the request.*
4. *For an approved request, staff records the blood issue.*
5. *The system updates the request status.*
6. *The issued blood is deducted from the inventory.*
7. *The system records the completed transaction.*

5.7.3 Functional Requirements

FR-7.1: *The system shall allow authorized blood bank staff to approve or reject a pending blood request.*

FR-7.2: *The system shall allow blood to be issued only against an approved request.*

FR-7.3: *The system shall verify sufficient available inventory before blood is issued.*

FR-7.4: *The system shall prevent issuance when sufficient blood is unavailable.*

FR-7.5: *The system shall reduce the corresponding blood inventory after a successful issue.*

FR-7.6: *The system shall update the request status after blood has been issued.*

FR-7.7: *The system shall maintain a record of each blood issue.*

5.8 Reports and History

5.8.1 Description and Priority

This feature provides authorized users with access to historical records and basic reports related to blood bank operations.

Priority: Medium

5.8.2 Stimulus/Response Sequences

1. *An authorized user opens the reports or history section.*
2. *The user selects the required type of information.*
3. *The system retrieves the relevant records.*
4. *The system displays the requested information.*

5.8.3 Functional Requirements

FR-8.1: *The system shall maintain a history of recorded blood donations.*

FR-8.2: *The system shall maintain a history of blood requests and their status.*

FR-8.3: *The system shall maintain a history of blood issued against requests.*

FR-8.4: *The system shall allow authorized users to view current blood inventory information.*

FR-8.5: The system shall allow authorized users to view basic reports related to donors, donations, inventory, requests, and blood issuance.

5.9 User Authentication and Management

5.9.1 Description and Priority

This feature allows authorized users to authenticate themselves before accessing protected system functions. It also allows administrators to create and manage authorized user accounts and assign roles that determine access to system functions.

Priority: High

5.9.2 Stimulus/Response Sequences

1. The user opens the login screen.
2. The user enters valid login credentials.
3. The system verifies the credentials and authenticates the user.
4. The system provides access to functions permitted for the user's assigned role.
5. If the credentials are invalid, the system rejects the login and displays an appropriate message.
6. An authenticated user may log out of the system.
7. On logout, the system invalidates the user's active session.
8. An administrator may create authorized user accounts and assign or modify user roles.

5.9.3 Functional Requirements

FR-9.1: The system shall allow authorized users to log in using valid credentials.

FR-9.2: The system shall reject invalid login credentials and display an appropriate message.

FR-9.3: The system shall allow authenticated users to log out.

FR-9.4: The system shall invalidate the active user session after logout.

FR-9.5: The system shall allow an Administrator to create authorized user accounts.

FR-9.6: The system shall allow an Administrator to assign and modify user roles.

FR-9.7: The system shall restrict access to system functions according to the user's assigned role.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

NFR-01: Donor search, blood availability search and inventory view shall respond within 3 seconds.

NFR-02: Submission of donations, requests and issues shall be processed within 3 seconds.

NFR-03: Reports shall display within 5 seconds.

NFR-04: Inventory changes shall be reflected immediately after a successful transaction.

6.2 Safety Requirements

Because the system manages information related to blood availability and issuance, incorrect records could affect blood-bank operations. The system shall therefore include safeguards against invalid or unsafe operations.

SAFE-01: Expired blood units shall not be shown as available for issue.

SAFE-02: Blood shall be issued only against an approved blood request.

SAFE-03: The system shall prevent blood from being issued when sufficient stock is not available.

SAFE-04: The system shall require confirmation before completing a blood-issue operation that changes inventory.

SAFE-05: Medical screening and final suitability of donated blood shall remain the responsibility of authorized blood-bank personnel and shall not be determined by the software.

6.3 Security Requirements

Security Objectives

SO-01 - Confidentiality:

Prevent unauthorized access to donor information, user information, blood requests, and other protected system records.

SO-02 - Integrity:

Protect donation, inventory, request, and blood-issue records from unauthorized modification.

SO-03 - Access Control:

Ensure that users can access only the functions permitted for their assigned role.

Security Requirements

SEC-01: *The system shall require users to authenticate before accessing protected functions.*

SEC-02: *The system shall provide role-based access to system functions.*

SEC-03: *Only authorized blood-bank staff shall be permitted to modify blood inventory or record blood issuance.*

SEC-04: *Administrative functions such as user management shall be restricted to administrator accounts.*

SEC-05: *Passwords shall be masked during entry and shall be stored only as salted cryptographic hashes.*

SEC-06: *Communication between the browser and deployed application shall use HTTPS with TLS 1.2 or higher.*

SEC-07: *The system shall protect state-changing requests against Cross-Site Request Forgery (CSRF).*

SEC-08: *The system shall validate and safely process user input to prevent SQL injection attacks.*

SEC-09: *The system shall escape or sanitize user-supplied content to prevent Cross-Site Scripting (XSS).*

SEC-10: *The system shall invalidate the authenticated session when a user logs out.*

6.4 Software Quality Attributes

NFR-05 Usability: A trained staff user shall be able to record a donation within 2 minutes.

NFR-06 Reliability: A failed transaction shall leave inventory unchanged.

NFR-07 Maintainability: Each functional area shall be a separate Django app.

NFR-08 Portability: The app shall work on the latest Chrome, Edge and Firefox on Windows, macOS and Linux.

NFR-09 Testability: Every FR shall have at least one test case.

6.5 Business Rules

BR-01: *Only authorized blood-bank staff may register a completed blood donation.*

BR-02: *Only blood that has been accepted by authorized staff may be added to available inventory.*

BR-03: *Expired blood units shall not be considered part of available stock.*

BR-04: *A blood request must be approved before blood can be issued against it.*

BR-05: *Blood shall not be issued if the required quantity is unavailable.*

BR-06: *Every blood issue shall be associated with a corresponding blood request.*

BR-07: *Issuing blood shall reduce the available quantity of the corresponding blood group.*

BR-08: *Each donation shall be associated with a registered donor.*

6.6 Domain Requirements

DR-01: *The system shall support the standard blood groups A+, A-, B+, B-, AB+, AB-, O+, and O-.*

DR-02: *Each donation record shall be associated with a blood group and a recorded quantity or number of units.*

DR-03: *Blood inventory shall be maintained separately according to blood group.*

DR-04: *Each blood request shall specify the required blood group and quantity.*

DR-05: *Donation, request, and issue records shall include the relevant date and status information.*

7. Other Requirements

7.1 Database Requirements

The system shall use a relational database to maintain information related to users, donors, donations, blood inventory, blood requests, and blood issuance.

Each record shall have a unique identifier where required. Relationships between records shall be maintained so that, for example, donations can be traced to donors and blood issues can be traced to their corresponding requests.

The database shall maintain consistency between blood inventory and transactions that add or remove blood units.

7.2 Future Scope

Future versions of the system may provide a donor portal that allows donors to create accounts, view donation history, perform a preliminary donation eligibility check, and locate nearby participating blood banks or donation centres.

Appendix A: Glossary

Term / Abbreviation	Meaning
SRS	<i>Software Requirements Specification. A document that defines the functional and non-functional requirements of the system.</i>
FR	<i>Functional Requirement. A requirement that describes a function or service the system must provide.</i>
NFR	<i>Non-Functional Requirement. A requirement related to qualities such as performance, security, reliability, or usability.</i>
Donor	<i>A person who donates blood to the blood bank.</i>
Donation	<i>A recorded instance of blood being donated by a registered donor.</i>
Blood Unit	<i>A stored quantity of donated blood recorded and managed by the blood bank.</i>
Blood Inventory	<i>The current stock of available blood maintained according to blood group and status.</i>
Blood Request	<i>A request submitted by an authorized hospital or requesting organization for a required blood group and quantity.</i>
Blood Issue	<i>The process of providing blood against an approved request and updating the available inventory.</i>

Blood Bank Staff	<i>Authorized personnel responsible for donor registration, donation recording, inventory management, request processing, and blood issuance.</i>
Requesting Organization	<i>A hospital or other authorized organization that searches for blood availability and submits blood requests.</i>
Administrator	<i>A user with administrative privileges for accessing reports and managing authorized system functions.</i>
Expired Blood Unit	<i>A blood unit that has passed its recorded expiry date and must not be considered available for issue.</i>
Low Stock	<i>A condition in which the available quantity of a blood group falls below 5 units.</i>
HTTP	<i>Hypertext Transfer Protocol, used for communication between a web browser and server.</i>
HTTPS	<i>Secure version of HTTP used to protect data transmitted between the browser and server.</i>
Django	<i>Python-based web framework used to implement the application backend.</i>
RBAC	<i>Role-Based Access Control. A method of restricting system functions according to the role assigned to a user.</i>

Appendix B: Field Layouts

B.1 Donor Registration

Field	Length	Data Type	Description	Is Mandatory
Donor ID	10	Alphanumeric	Unique identifier for the donor	Y
Full Name	100	String	Name of the donor	Y
Date of Birth	8	Date	Donor's date of birth	Y
Blood Group	3	String	A+, A-, B+, B-, AB+, AB-, O+, O-	Y
Phone Number	15	String	Contact number of the donor	Y
Email	100	String	Donor's email address	N

Address	255	String	Residential address	N
Registration Date	8	Date	Date on which donor was registered	Y
Status	20	String	Current donor record status	Y

B.2 Donation Recording

Field	Length	Data Type	Description	Is Mandatory
Donation ID	10	Alphanumeric	Unique identifier for the donation	Y
Donor ID	10	Alphanumeric	Donor associated with the donation	Y
Donation Date	8	Date	Date of donation	Y
Blood Group	3	String	Blood group of donated blood	Y
Quantity / Units	-	Numeric	Quantity of blood donated	Y
Expiry Date	8	Date	Expiry date of the stored blood unit	Y
Donation Status	20	String	Pending, Accepted, Rejected	Y

B.3 Blood Request

Field	Length	Data Type	Description	Is Mandatory
Request ID	10	Alphanumeric	Unique identifier for the request	Y
Organization Name	100	String	Hospital or requesting organization	Y
Blood Group	3	String	Required blood group	Y
Quantity Required	-	Numeric	Number of units requested	Y
Request Date	8	Date	Date on which request was submitted	Y

Request Status	20	String	Pending, Approved, Rejected, Fulfilled	Y
Remarks	255	String	Additional information about the request	N

B.4 Blood Issue

Field	Length	Data Type	Description	Is Mandatory
Issue ID	10	Alphanumeric	Unique identifier for the blood issue	Y
Request ID	10	Alphanumeric	Approved request against which blood is issued	Y
Blood Group	3	String	Blood group issued	Y
Quantity Issued	-	Numeric	Number of units issued	Y
Issue Date	8	Date	Date blood was issued	Y
Issued By	50	String	Authorized staff responsible for issue	Y

B.5 Inventory Record

Field	Length	Data Type	Description	Is Mandatory
Blood Unit ID	10	Alphanumeric	Unique identifier for a stored blood unit	Y
Blood Group	3	String	Blood group of the unit	Y
Quantity	-	Numeric	Available quantity	Y
Collection Date	8	Date	Date the blood was collected	Y
Expiry Date	8	Date	Expiry date of the blood unit	Y
Status	20	String	Available, Issued, Expired	Y

Sample Report Requirements

Donor & Donation Report

Donor ID
Donor Name
Blood Group
Donation ID
Donation Date
Quantity / Units
Donation Status

Blood Inventory Report

Blood Unit ID
Blood Group
Available Quantity
Collection Date
Expiry Date
Status

Blood Request & Issue Report

Request ID
Requesting Organization
Blood Group
Quantity Requested
Request Date
Request Status
Issue ID
Quantity Issued
Issue Date

Appendix C: Requirement Traceability Matrix

Sl. No.	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-1.1- FR-1.3	Register donor, validate mandatory details, and store blood group	TBD - SAD	UC-02 Register Donor	TBD - Implementation	TC-DON-01 TC-DON-02	TBD
2	FR-1.4, FR-1.5	Search and update donor details	TBD - SAD	UC-03 Maintain Donor Details	TBD - Implementation	TC-DON-03	TBD
3	FR-1.6	Display donor donation history	TBD - SAD	UC-04 View Donation History	TBD - Implementation	TC-DNT-01	TBD

4	FR-1.7	Display validation messages for invalid donor data	TBD - SAD	UC-02 Register Donor	TBD - Implementation	TC-DON-02	TBD
5	FR-2.1 - FR-2.6	Record donation details and maintain donation history	TBD - SAD	UC-05 Record Donation	TBD - Implementation	TC-DNT-01	TBD
6	FR-2.7	Reject invalid donation data	TBD - SAD	UC-05 Record Donation	TBD - Implementation	TC-DNT-02	TBD
7	FR-3.1, FR-3.2, FR-3.4	Maintain, add, and view blood inventory	TBD - SAD	UC-06 Manage Blood Inventory	TBD - Implementation	TC-INV-01	TBD
8	FR-3.3	Reduce inventory when blood is issued	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-01	TBD
9	FR-3.5	Prevent inventory from becoming negative	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-03	TBD
10	FR-4.1, FR-4.2, FR-4.5	Track expired units and exclude them from issue	TBD - SAD	UC-07 Monitor Expiry & Low Stock	TBD - Implementation	TC-MON-01	TBD
11	FR-4.3, FR-4.4	Detect near-expiry units and low stock	TBD - SAD	UC-07 Monitor Expiry & Low Stock	TBD - Implementation	TC-MON-02	TBD

12	FR-5.1, FR-5.2	Search by blood group and display available quantity	TBD - SAD	UC-08 Search Blood Availability	TBD - Implementation	TC-SRCH-01	TBD
13	FR-5.3	Exclude expired units from availability results	TBD - SAD	UC-08 Search Blood Availability	TBD - Implementation	TC-SRCH-01 TC-MON-01	TBD
14	FR-5.4	Indicate when the requested blood group is unavailable	TBD - SAD	UC-08 Search Blood Availability	TBD - Implementation	TC-SRCH-02	TBD
15	FR-6.1 - FR-6.6	Submit, review, and track blood requests	TBD - SAD	UC-09 Create Blood Request; UC-10 Track Request Status; UC-11 Approve / Reject Request	TBD - Implementation	TC-REQ-01	TBD
16	FR-6.7	Reject incomplete blood requests	TBD - SAD	UC-09 Create Blood Request	TBD - Implementation	TC-REQ-02	TBD
17	FR-7.1	Approve or reject a pending blood request	TBD - SAD	UC-11 Approve / Reject Blood Request	TBD - Implementation	TC-ISS-01, TC-ISS-02	TBD
18	FR-7.2	Issue blood only against an approved request	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-02	TBD

19	FR-7.3, FR-7.4	Verify sufficient stock and prevent invalid issue	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-03	TBD
20	FR-7.5 - FR-7.7	Update inventory, request status, and blood-issue record	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-01	TBD
21	FR-8.1 - FR-8.5	Maintain and display reports and history	TBD - SAD	UC-13 View Reports & History	TBD - Implementation	TC-RPT-01	TBD
22	FR-9.1, FR-9.2	User login and invalid login handling	TBD - SAD	UC-01 Login	TBD - Implementation	TC-SEC-01	TBD
23	FR-9.3, FR-9.4	User logout and session invalidation	TBD - SAD	UC-01 Login	TBD - Implementation	TC-SEC-01	TBD
24	FR-9.5 - FR-9.7	User creation, role assignment, and role-based access	TBD - SAD	TBD - User Management Design / RBAC	TBD - Implementation	TC-SEC-02, TC-SEC-03	TBD
25	NFR-01	Search and inventory operations respond within 3 seconds	TBD - SAD	TBD - SAD	TBD - Implementation	TC-PERF-01	TBD
26	NFR-02	Donation, request, and issue submissions complete within 3 seconds	TBD - SAD	TBD - SAD	TBD - Implementation	TC-PERF-02	TBD

27	NFR-03	Reports display within 5 seconds	TBD - SAD	TBD - SAD	TBD - Implementation	TC-PERF-03	TBD
28	NFR-04	Inventory changes are reflected immediately	TBD - SAD	TBD - SAD	TBD - Implementation	TC-INV-01	TBD
29	NFR-05	Trained staff can record a donation within 2 minutes	TBD - SAD	TBD - SAD	TBD - Implementation	TC-USE-01	TBD
30	NFR-06	Failed transactions leave inventory unchanged	TBD - SAD	TBD - SAD	TBD - Implementation	TC-REL-01	TBD
31	NFR-07	Functional areas implemented as separate Django apps	TBD - SAD	TBD - SAD	TBD - Implementation	Design / Code Review	N/A
32	NFR-08	Application works across supported browsers and operating systems	TBD - SAD	TBD - SAD	TBD - Implementation	TC-PORT-01	TBD
33	NFR-09	Every functional requirement has at least one test case	N/A	RTM	N/A	Verified by RTM	N/A
34	SEC-01	Authentication required before accessing protected functions	TBD - SAD	UC-01 Login	TBD - Implementation	TC-SEC-01	TBD

35	SEC-02, SEC-03	Role-based access and staff-only inventory/issue operations	TBD - SAD	TBD - SAD (RBAC)	TBD - Implementation	TC-SEC-02	TBD
36	SEC-04	User-management functions restricted to Administrator	TBD - SAD	TBD - User Management Design	TBD - Implementation	TC-SEC-03	TBD
37	SEC-05	Passwords are protected and not displayed in plain text	TBD - SAD	UC-01 Login	TBD - Implementation	TC-SEC-04	TBD
38	SEC-06	Use HTTPS for browser-server communication	TBD - SAD	TBD - Security / Deployment Design	TBD - Implementation	TC-SEC-05	TBD
39	SEC-07	Protect state-changing requests against CSRF	TBD - SAD	TBD - Security Design	TBD - Implementation	TC-SEC-06	TBD
40	SEC-08	Prevent SQL injection through user input	TBD - SAD	TBD - Security Design	TBD - Implementation	TC-SEC-06	TBD
41	SEC-09	Prevent Cross-Site Scripting (XSS)	TBD - SAD	TBD - Security Design	TBD - Implementation	TC-SEC-06	TBD
42	SEC-10	Invalidate authenticated session on logout	TBD - SAD	UC-01 Login	TBD - Implementation	TC-SEC-01	TBD

43	SAFE-01	Expired blood units shall not be available for issue	TBD - SAD	UC-07 Monitor Expiry & Low Stock	TBD - Implementation	TC-MON-01	TBD
44	SAFE-02	Blood shall only be issued against an approved request	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-02	TBD
45	SAFE-03	Prevent blood issue when stock is insufficient	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-03	TBD
46	SAFE-04	Require confirmation before completing blood issue	TBD - SAD	UC-12 Issue Blood	TBD - Implementation	TC-ISS-01	TBD
47	SAFE-05	Medical screening and suitability remain outside the software	N/A	N/A	N/A	Not Tested - Out of Scope	N/A